Chapter 5, Problem 17P

Problem

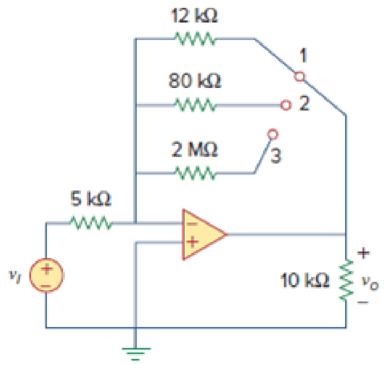
Calculate the gain v_o/v_i when the switch in Fig. 5.56 is in:

(a) position 1

(b) position 2

(c) position 3.

Figure. 5.56:



Step-by-step solution

Step 1 of 6

(a)

Refer to Figure 5.56P in the textbook.

Consider the switch is in position 1.

Draw the following Figure 1.

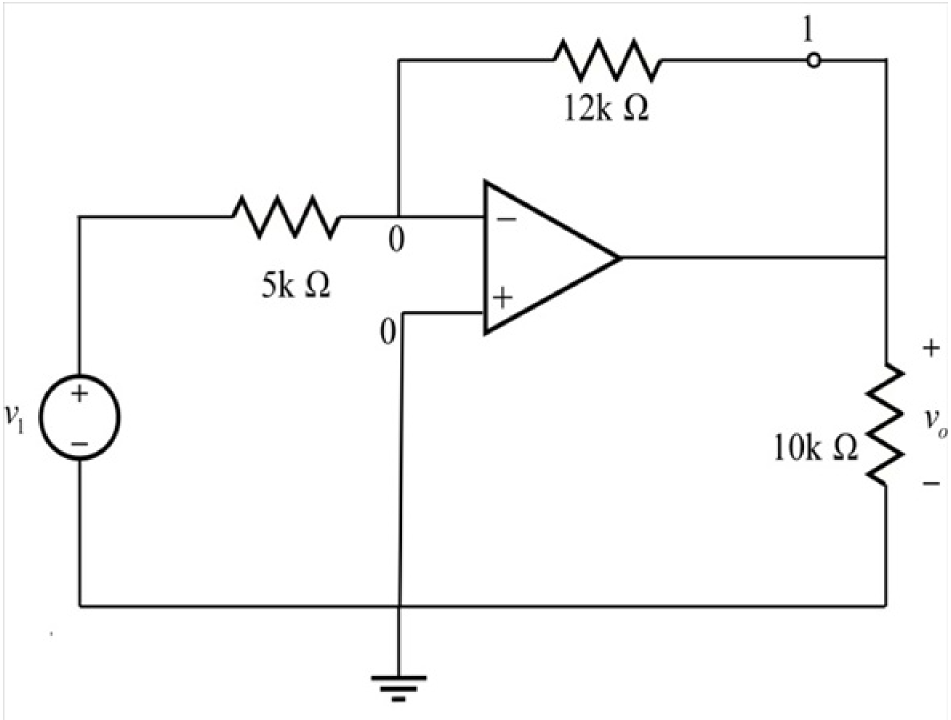


Figure 1

Step 2 of 6

Determine the ratio of $\frac{v_o}{v_i}$.

Apply Kirchhoff's current law at inverting terminal.

$$\frac{0-v_i}{5\text{ k}\Omega}+\frac{0-v_o}{12\text{ k}\Omega}=0$$

$$-\frac{v_i}{5\text{ k}\Omega}+\frac{-v_o}{12\text{ k}\Omega}=0$$

$$-\frac{v_i}{5\text{ k}\Omega}=\frac{v_o}{12\text{ k}\Omega}$$

$$v_o=-\frac{12}{5}v_i$$

$$v_o=-2.4v_i$$

$$\frac{v_o}{v_i}=-2.4$$

Therefore, the ratio of $\frac{v_o}{v_i}$ in position 1 is $\boxed{-2.4}$.

Step 3 of 6

(b)

Consider the switch is in position 2.

Draw the modified circuit as shown in Figure 2.

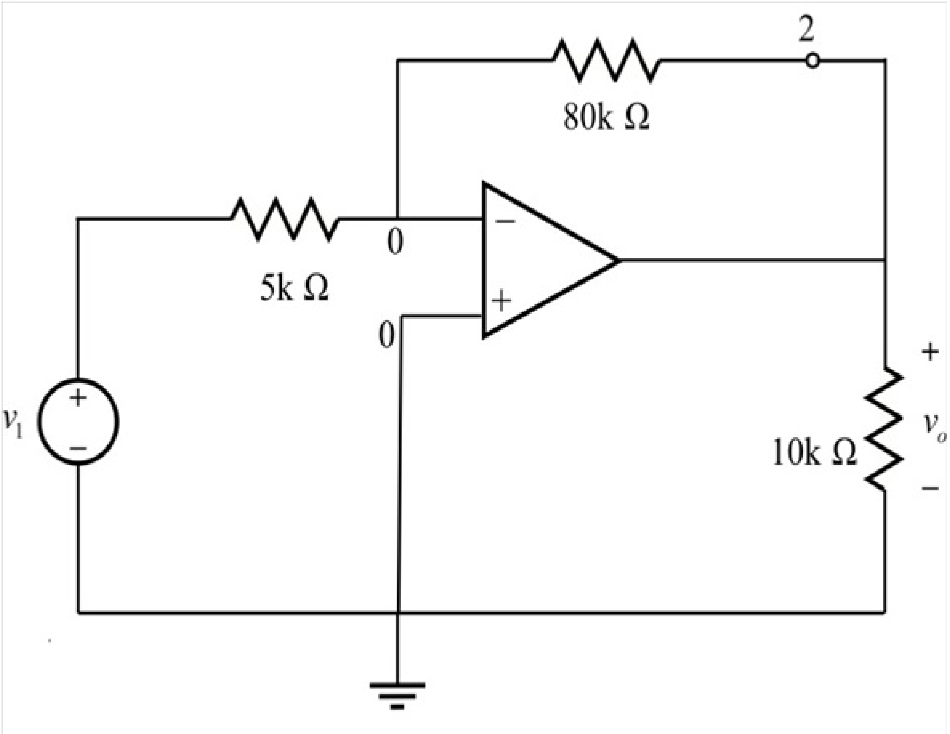


Figure 2

Step 4 of 6

Determine the ratio of $\frac{v_o}{v_i}$.

Apply Kirchhoff's current law at inverting terminal.

$$\frac{0-v_i}{5\ \text{k}\Omega}+\frac{0-v_o}{80\ \text{k}\Omega}=0$$

$$-\frac{v_i}{5\ \text{k}\Omega}+\frac{-v_o}{80\ \text{k}\Omega}=0$$

$$-\frac{v_i}{5\ \text{k}\Omega}=\frac{v_o}{80\ \text{k}\Omega}$$

$$v_o=-\frac{80}{5}v_i$$

$$v_o=-16v_i$$

$$\frac{v_o}{v_i}=-16$$

Therefore, the ratio of $\frac{v_o}{v_i}$ in position 2 is $\boxed{-16}$.

Step 5 of 6

(c)

Consider the switch is in position 3.

Draw the modified circuit as shown in Figure 3.

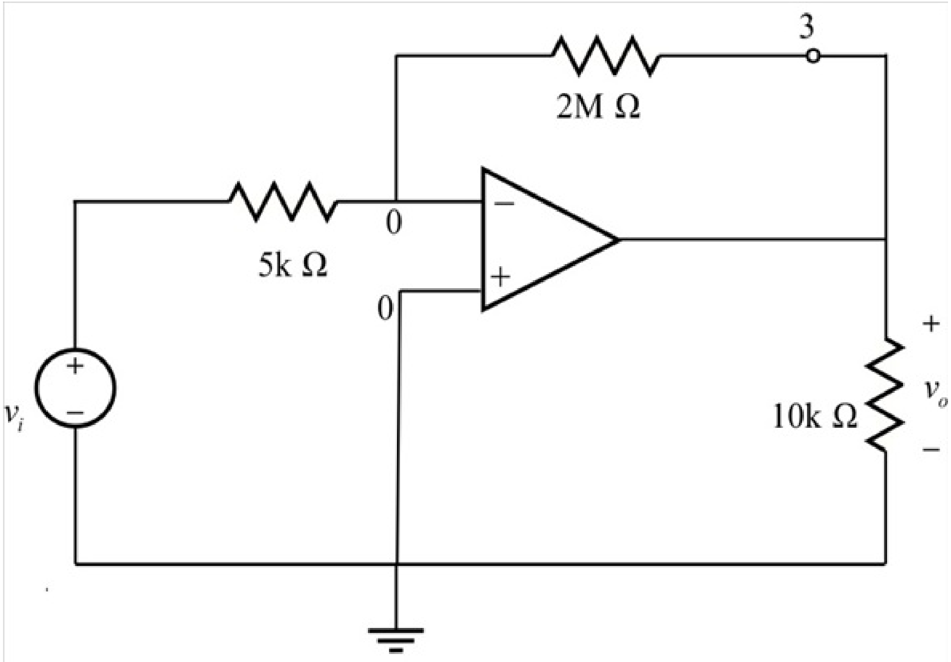


Figure 3

Step 6 of 6

Determine the ratio of $\frac{v_o}{v_i}$.

Apply Kirchhoff's current law at inverting terminal.

$$\frac{0-v_i}{5\ \text{k}\Omega}+\frac{0-v_o}{2000\ \text{k}\Omega}=0$$

$$-\frac{v_i}{5\ \text{k}\Omega}+\frac{-v_o}{2000\ \text{k}\Omega}=0$$

$$-\frac{v_i}{5\ \text{k}\Omega}=\frac{v_o}{2000\ \text{k}\Omega}$$

$$v_o=-\frac{2000}{5}v_i$$

$$v_o=-400v_i$$

$$\frac{v_o}{v_i}=-400$$

Therefore, the ratio of $\frac{v_o}{v_i}$ in position 3 is $\boxed{-400}$.